

Minimizing total treatment time

The problem is

$$\min_{x_1, x_2 \geq 0} x_1 + x_2, \quad \text{s.t. } x_1 + a_2 x_2 = 1,$$

with $a_1=1$. The feasible set is the line segment from $(x_1, x_2)=(1,0)$ to $(0, 1/a_2)$. By linear-programming theory, an optimal solution occurs at an extreme (vertex) of this segment ¹. Evaluating the two endpoints:

- At $(1,0)$ the total time is $1+0=1$.
- At $(0, 1/a_2)$ the total time is $0 + 1/a_2 = 1/a_2$.

Thus the better endpoint is the one with smaller value. Since a_2 (as defined by the code) turns out to exceed 1 , we have $1/a_2 < 1$ and the optimal minimiser is $(x_1, x_2)=(0, 1/a_2)$.

From the code we see that a_2 is given by

$$a_2 = 2 \lim_{n \rightarrow \infty} f(n) + \frac{1}{2}, \quad f(n) = 72^{-10^{24}} + \sum_{m=1}^n \left(2^{-k} - 72^{-10^{24}k} \right), \quad k = 2 + m.$$

In the limit $n \rightarrow \infty$, the main contribution is $\sum_{k=3}^{\infty} 2^{-k} = 0.25$, so $2f(\infty) + 0.5 = 1$ up to tiny corrections. In fact one finds

$$a_2 = 1 + 2 \cdot 72^{-10^{24}} - 2 \sum_{k=3}^{\infty} 72^{-10^{24}k},$$

i.e. $a_2 = 1 +$ an *exponentially* small positive amount. (By definition a **computable number** is one given by an algorithm like this, which can be calculated to arbitrarily high precision ².) Hence $a_2 > 1$, but a_2 differs from 1 by far less than 10^{-24} , so to all practical accuracy $a_2 \approx 1.000 \dots$.

Therefore the optimal solution is

$$x_1 = 0, \quad x_2 = 1/a_2.$$

Numerically $a_2 \approx 1$, so $x_2 = 1/a_2 \approx 1.00$ to within 10^{-2} . In other words, to two-decimal accuracy the minimiser is

$$(x_1, x_2) \approx (0.00, 1.00).$$

(The ℓ^∞ error is ≈ 0.01 since $|1 - 1/a_2| \approx 2 \cdot 72^{-10^{24}}$ is essentially zero at that scale.)

Answer: $x_1 \approx 0.00, x_2 \approx 1.00$ (to two-decimal accuracy).

Sources: Extreme-point optimality in LP ¹; definition of computable real numbers ².

¹ feog.github.io

<https://feog.github.io/chap4dm.pdf>

² Alan Turing and the Countability of Computable Numbers – Feature Column

<https://mathvoices.ams.org/featurecolumn/2021/12/01/alan-turing-computable-numbers/>